



PROJECT REPORT

Roadmap Tracker for Career Goals

1. Introduction

In today's fast-paced learning environment, students and professionals often struggle to track their learning progress effectively. Many people set goals but fail to achieve them due to lack of proper tracking and motivation.

The Roadmap Tracker is a web application designed to solve this problem. It helps users:

- Create and manage learning goals
- Visualize their progress through interactive roadmaps
- Stay motivated with daily streaks and achievement badges
- Track completion status with visual progress indicators

The application is built as a Progressive Web App (PWA) that works offline and saves all data locally in the user's browser, ensuring privacy and accessibility without requiring an internet connection or backend server.

2. Abstract

Career Roadmap Tracker is a React-based web application that enables users to plan, track, and visualize their learning journey. The app provides four main features:

1. Goal Management – Add, edit, delete, and track progress of learning goals
2. Visual Roadmap – Generate dynamic flowcharts showing goal completion status
3. Streak Tracker – Maintain daily check-ins to build consistent learning habits
4. Achievement Badges – Earn rewards for completing milestones

All data is stored locally using the browser's localStorage API, making the app fully offline-capable and privacy-focused. The application features a modern glassmorphism UI design, dark/light mode toggle, and smooth animations for an engaging user experience.

Key outcomes:

- Users can track unlimited learning goals
- Visual representation of progress motivates consistent effort
- Gamification elements (streaks + badges) increase engagement
- No backend required – works completely offline

3. Tools & Technologies Used

Category	Technology	Purpose
Frontend Framework	React.js (v18)	Building user interface components
Styling	Tailwind CSS + Custom CSS	Modern responsive design
Diagrams	Mermaid.js	Generating visual roadmap flowcharts
Data Storage	localStorage (Browser API)	Saving goals and progress locally
Icons	Emoji / Unicode	Visual enhancements
Development	Create React App	Project scaffolding
Version Control	Git	Code management

Development Environment:

- Code Editor: VS Code
- Browser: Chrome/Firefox/Edge
- Operating System: Windows/Mac/Linux compatible

4. Steps Involved in Building the Project

Step 1: Project Setup

- Created React app using `npx create-react-app roadmap-tracker`
- Installed required packages: `mermaid`, `tailwindcss`
- Configured Tailwind CSS for styling

Step 2: Component Architecture

Designed and built 6 reusable components:

Component	Function
<code>App.js</code>	Main application container, state management
<code>GoalForm.js</code>	Form to add new goals with priority & deadline
<code>GoalList.js</code>	Display all goals with progress sliders
<code>RoadmapChart.js</code>	Generate Mermaid.js flowchart diagram
<code>StreakTracker.js</code>	Daily check-in system with streak counting
<code>Badges.js</code>	Achievement system based on milestones

Step 3: State Management

- Used React `useState` hooks for local component state
- Used React `useEffect` hooks for `localStorage` sync
- Implemented lift-state-up pattern for data flow

Step 4: Data Persistence

- Implemented localStorage CRUD operations:
 - Create: Add new goals with unique ID
 - Read: Load saved goals on app startup
 - Update: Modify progress percentage (0-100%)
 - Delete: Remove unwanted goals

Step 5: Visual Roadmap Integration

- Integrated Mermaid.js library
- Created dynamic flowchart based on:
 - Total goals count
 - Completed vs in-progress vs not started
 - Visual styling with colors

Step 6: Gamification Features

- Streak System: Check-in daily, resets if missed
- Badge System: Unlocks at milestones (first goal, 5 goals, 100% completion)
- Confetti Effect: Triggers when goal reaches 100%

Step 7: UI/UX Enhancement

- Glassmorphism design with backdrop blur
- Dark/Light mode toggle with localStorage persistence
- Responsive layout for mobile/tablet/desktop
- Smooth hover animations and transitions

Step 8: Testing & Debugging

- Tested all CRUD operations
- Verified localStorage persistence after page refresh
- Tested responsive design on different screen sizes
- Fixed compilation errors (missing component imports)

Step 9: Final Polish

- Added footer with data privacy note
- Implemented scrollbar styling
- Added loading states and empty state messagesA

5. Conclusion

The Career Roadmap Tracker project was successfully completed within the 2-day timeframe. The application meets all the requirements of Project #14 from the project guidelines:

✅ Core Features Implemented:

- Add/edit learning goals
- Visualize roadmaps with Mermaid.js
- Store progress in localStorage
- Streak tracking for motivation
- Badge system for achievements

✅ Additional Enhancements:

- Dark/Light mode toggle
- Priority levels (High/Medium/Low)
- Deadline picker for goals
- Confetti animation on completion
- Fully responsive design
- Glassmorphism UI effects

Learning Outcomes:

- Gained practical experience with React hooks (useState, useEffect)
- Learned to integrate third-party libraries (Mermaid.js)
- Understood browser storage mechanisms (localStorage)
- Improved UI/UX design skills with Tailwind CSS
- Learned to build offline-first web applications

Future Scope:

- Add cloud sync using Firebase
- Implement email reminders for goals
- Add social sharing of roadmaps
- Export roadmap as PDF/Image
- Add collaborative features (shared roadmaps)

Project Status: ✅ COMPLETED

Date: May 12, 2026

